2.6.4. Comprehensions

Comprehensions provide a way to look at the array of things (such as ingredients in a recipe) and to manipulate the values without having to use loops. CoffeeScript provides a rich set of comprehensions that can apply to either arrays or objects.

for...in... comprehension

Array comprehensions allow you to evaluate an expression for each item in an array. Here's a one-line comprehension that's easy to experiment with on the REPL:

```
number for number in [9,0,2,1,0]
# [9,0,2,1,0]
```

Using the name `number` in the comprehension declares it as a variable. You can use any variable name you like:

```
x for x in [9,0,2,1,0]
# [9,0,2,1,0]
```

However, it is best to use a different variable name just for the constructor (you'll learn more about why later on). Now use a comprehension to add 1 to every item in the array:

```
number + 1 for number in [9,0,2,1,0]
# [10,1,3,2,1]
```

Use a comprehension to convert every item to a 0:

```
0 for number in [9,0,2,1,0]
# [0,0,0,0,0]
```

The name after the `for` keyword in a comprehension creates a variable with that name. It's possible to access the variable outside of the comprehension:

```
letter for letter in ['x','y','z']
# [x,y,z]
```